Properties of the Lebesgue integral

REAL VARIABLES I

Fall 2026

In the previous note, we constructed the Lebesgue integral for non-negative measurable functions and proved the monotone convergence theorem, which turns pointwise approximation from below into convergence of integrals. In these notes, we first apply monotone convergence to compute the integral of an unbounded function directly, demonstrating that the Lebesgue theory handles singularities without needing a separate framework of improper integrals. We then investigate the structural consequences of having a finite integral. Markov's inequality provides quantitative control on the measure of superlevel sets, implying in particular that an integrable function is finite almost everywhere. Next, we examine uniform integrability, which shows that a function with finite integral cannot concentrate a positive amount of integral on sets of arbitrarily small measure. Finally, we decompose signed functions into their positive and negative parts to define the Lebesgue integral for general real-valued functions and establish its linearity.

After studying these notes, you will be able to do the following.

- Compute an unbounded Lebesgue integral by superlevel sets and by monotone truncation.
- Apply Markov's inequality to estimate the measure of a superlevel set.
- Explain why every measurable function has integral zero over a null set, even when it takes the value $+\infty$ there.
- Prove that a non-negative function with finite integral has uniformly small integrals over sets of small measure.
- Define the integral of an integrable signed function and prove its linearity.

Throughout these notes, $(X, \mathcal{A}, \mu)$ denotes a measure space. Unless a signed function is explicitly under discussion, measurable functions take values in $[0, +\infty]$.

8.1 An unbounded function with finite integral

First, we will show that an unbounded function can still have a finite integral. On $\mathbb{R}$ with Lebesgue measure, let

$$f(x) \coloneqq \begin{cases} x^{-1/2}, & 0 < x \leq 1, \\ 0, & x \notin (0, 1]. \end{cases}$$

This function is also written $x^{-1/2}\mathbb{1}_{(0,1]}(x)$; the value outside $(0, 1]$ set to zero. It is measurable because the identity and the indicator of $(0, 1]$ are measurable, and products, continuous compositions, and quotients with nonzero denominator preserve measurability on $(0, 1]$; extension by zero across a measurable complement preserves measurability.

We will compute the integral in two ways.

Example 8.1 (Computation by superlevel sets). For $0 \leq \lambda < 1$, the superlevel set $\{x \in \mathbb{R} : f(x) > \lambda\}$ is $(0, 1]$ and has Lebesgue measure 1. For $\lambda \geq 1$,

$$\{x \in \mathbb{R} : f(x) > \lambda\} = (0, \lambda^{-2}),$$

which has Lebesgue measure λ^{-2}. The layer-cake definition and the Archimedean integral therefore give

$$\int_{\mathbb{R}} f(x)\ \mathrm{d}x = \int_0^1 1\ \mathrm{d}\lambda + \int_1^{+\infty} \frac{1}{\lambda^2}\ \mathrm{d}\lambda = 1 + \left[-\frac{1}{\lambda}\right]_1^{+\infty} = 2.$$

The second integral was evaluated as an improper Riemann integral (which coincides with the Archimedean integral for finite-valued functions). This shows that the singularity at 0 of $x^{-1/2}\mathbb{1}_{(0,1]}$ does not make the integral infinite.

Example 8.2 (Computation by monotone truncation). Multiplying a function by an indicator $\mathbb{1}_E$ restricts its support to E while setting it to zero on $X \setminus E$. For any measurable subset $E \subseteq X$, integrating over E is defined by integrating the product $f\mathbb{1}_E$ over the ambient space:

$$\int_E f\ \mathrm{d}\mu \coloneqq \int_X f\mathbb{1}_E\ \mathrm{d}\mu.$$

To evaluate $\int_{\mathbb{R}} f(x)\ \mathrm{d}x$, we truncate the domain of integration away from the singularity at 0. For each $n \geq 1$, let $E_n \coloneqq [1/n, 1]$ and define

$$f_n(x) \coloneqq f(x)\mathbb{1}_{[1/n,1]}(x) = x^{-1/2}\mathbb{1}_{[1/n,1]}(x).$$

Let us verify the hypotheses of the monotone convergence theorem for the sequence $(f_n)_{n\geq 1}$:

1. *Non-negativity and measurability:* Each f_n is a product of non-negative measurable functions, hence non-negative and measurable.
2. *Monotonicity:* Because $[1/n, 1] \subseteq [1/n+1, 1]$ for every $n \geq 1$, the indicator functions satisfy $\mathbb{1}_{[1/n,1]} \leq \mathbb{1}_{[1/n+1,1]}$. Since $f \geq 0$, it follows that $f_n(x) \leq f_{n+1}(x)$ for every $x \in \mathbb{R}$.
3. *Pointwise convergence $f_n(x) \uparrow f(x)$:* If $x \notin (0, 1]$, then $f(x) = 0$, and $x \notin [1/n, 1]$ for all $n \geq 1$, so $f_n(x) = 0 = f(x)$. If $x \in (0, 1]$, there exists an integer $N \in \mathbb{N}$ such that $1/N \leq x$. For all $n \geq N$, we have $x \in [1/n, 1]$, so $f_n(x) = x^{-1/2} = f(x)$. Hence $f_n(x) \uparrow f(x)$ for every $x \in \mathbb{R}$.

On each compact interval $[1/n, 1]$, the function $x \mapsto x^{-1/2}$ is continuous. Using the agreement between the Lebesgue and Riemann integrals for continuous functions on compact intervals (to be proved in Note 23),

$$\int_{\mathbb{R}} f_n(x)\ \mathrm{d}x = \int_{1/n}^1 \frac{1}{\sqrt{x}}\ \mathrm{d}x = \left[2\sqrt{x}\right]_{1/n}^1 = 2\left(1 - \frac{1}{\sqrt{n}}\right).$$

Applying the monotone convergence theorem to the increasing sequence $f_n \uparrow f$ yields

$$\int_{\mathbb{R}} f(x)\ \mathrm{d}x = \lim_{n\to\infty} \int_{\mathbb{R}} f_n(x)\ \mathrm{d}x = \lim_{n\to\infty} 2\left(1 - \frac{1}{\sqrt{n}}\right) = 2.$$

This method replaces an improper endpoint by an increasing exhaustion of the domain by ordinary compact intervals, on which the integral can be evaluated by standard calculus techniques.

8.2 Markov's inequality and null sets

If a non-negative function satisfies $f(x) \geq \lambda$ on a measurable set E, its integral is bounded below by the integral of the constant slice $\lambda\mathbb{1}_E$, which equals $\lambda\mu(E)$. This observation is the content of Markov's inequality, which bounds the measure of a superlevel set using the integral of the function.

Theorem 8.3 (Markov's inequality). Let $f\colon X \to [0,+\infty]$ be a measurable function. For every $\lambda \in [0,+\infty)$,

$$\mu(\{x \in X : f(x) \geq \lambda\}) \leq \frac{1}{\lambda}\int_X f \, \mathrm{d}\mu. \tag{8.1}$$

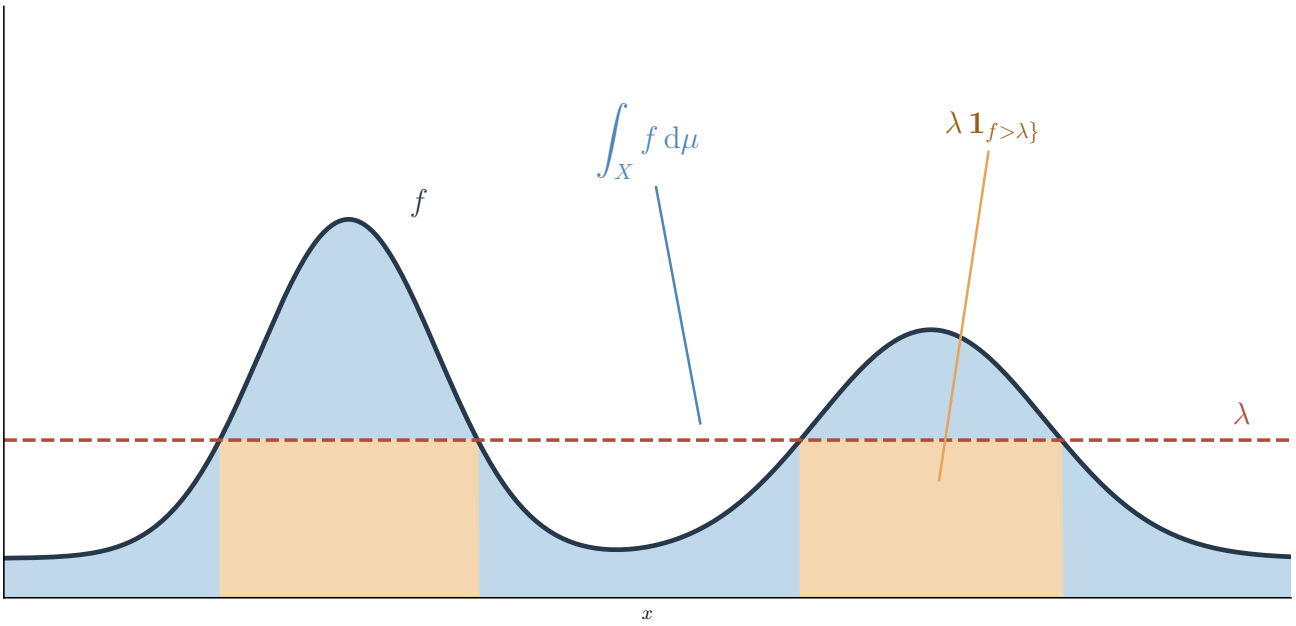


Figure 8.1: The blue area under f represents $\int_X f \, \mathrm{d}\mu$; the orange strip marks E_λ on the x-axis and colors the simple function $\lambda \mathbf{1}_{E_\lambda}$.

Figure description: A plot of a non-negative curve f with its area shaded blue. A horizontal level lambda marks the superlevel set E lambda in orange on the axis, and the orange region below the level represents the simple function lambda times the indicator of E lambda.

Proof of Theorem 8.3. The pointwise inequality $\lambda \mathbb{1}_{\{f \geq \lambda\}} \leq f$ holds everywhere on X. Monotonicity and positive homogeneity of the non-negative integral yield

$$\lambda\mu(\{f \geq \lambda\}) = \int_X \lambda \mathbb{1}_{\{f \geq \lambda\}} \, \mathrm{d}\mu \leq \int_X f \, \mathrm{d}\mu.$$

Using that $\lambda < +\infty$ to divide both sides by $\lambda > 0$ proves (8.1). □

Markov's inequality is especially informative when the integral $\int_X f \, \mathrm{d}\mu$ is known to be finite. In that case, letting $\lambda \to +\infty$ forces the measure of the superlevel set to vanish.

Corollary 8.4 (Finite integral implies finiteness almost everywhere). If $f\colon X \to [0,+\infty]$ is measurable and $\int_X f \, \mathrm{d}\mu < +\infty$, then

$$\mu(\{x \in X : f(x) = +\infty\}) = 0.$$

Proof of Corollary 8.4. For every $n \in \mathbb{N}$, the set $\{f = +\infty\}$ is contained in $\{f \geq n\}$. Applying Markov's inequality with $\lambda = n$ yields

$$\mu(\{f = +\infty\}) \leq \mu(\{f \geq n\}) \leq \frac{1}{n}\int_X f \, \mathrm{d}\mu.$$

Because $\int_X f \, \mathrm{d}\mu < +\infty$, the right-hand side converges to 0 as $n \to \infty$. Monotonicity of the measure forces

$\mu(\{f = +\infty\}) = 0$. □

Although a function with finite integral can only take the value $+\infty$ on a null set, the integral over any null set vanishes regardless of the values the function takes there.

Proposition 8.5 (Integral over a null set). Let $N \in \mathcal{A}$ satisfy $\mu(N) = 0$, and let $f\colon X \to [0, +\infty]$ be a measurable function. Then

$$\int_N f \, \mathrm{d}\mu \coloneqq \int_X f \mathbb{1}_N \, \mathrm{d}\mu = 0.$$

This conclusion remains valid even when $f(x) = +\infty$ for every $x \in N$.

Proof of Proposition 8.5. For every $\lambda > 0$, the superlevel set $\{x \in X : f(x)\mathbb{1}_N(x) > \lambda\}$ is contained in N. By monotonicity of the measure, $\mu(\{f\mathbb{1}_N > \lambda\}) \le \mu(N) = 0$, so $\mu(\{f\mathbb{1}_N > \lambda\}) = 0$ for every $\lambda > 0$. By the layer-cake formula,

$$\int_X f\mathbb{1}_N \, \mathrm{d}\mu = \int_0^{+\infty} \mu(\{f\mathbb{1}_N > \lambda\}) \, \mathrm{d}\lambda = \int_0^{+\infty} 0 \, \mathrm{d}\lambda = 0.$$

□

The following properties collect the standard consequences of Markov's inequality, Proposition 8.5, monotonicity, and positive linearity.

Proposition 8.6 (Fundamental almost-everywhere properties). Let $f, g\colon X \to [0, +\infty]$ be measurable functions.

1. $\int_X f \, \mathrm{d}\mu = 0$ if and only if $f = 0$ almost everywhere.
2. If $f \le g$ almost everywhere, then $\int_X f \, \mathrm{d}\mu \le \int_X g \, \mathrm{d}\mu$.
3. If $f = g$ almost everywhere, then $\int_X f \, \mathrm{d}\mu = \int_X g \, \mathrm{d}\mu$.
4. If $f \le g$ almost everywhere and $\mu(\{x \in X : f(x) < g(x)\}) > 0$, then $\int_X f \, \mathrm{d}\mu < \int_X g \, \mathrm{d}\mu$.

8.3 Uniform integrability

If a non-negative measurable function is bounded on X, say $0 \le f(x) \le M$ for all $x \in X$, then for every measurable set $A \in \mathcal{A}$,

$$\int_A f \, \mathrm{d}\mu \le \int_A M \, \mathrm{d}\mu = M\mu(A).$$

Consequently, whenever $\mu(A) < \varepsilon/M$, the integral satisfies $\int_A f \, \mathrm{d}\mu < \varepsilon$. When f is unbounded, no uniform bound M exists. Nevertheless, having a finite total integral $\int_X f \, \mathrm{d}\mu < +\infty$ guarantees the same qualitative behavior: the integral over A is forced to be arbitrarily small whenever $\mu(A)$ is sufficiently small.

Definition 8.7 (Uniform integrability over measurable sets). A measurable function $f\colon X \to [0, +\infty]$ is **uniformly integrable over measurable sets** if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$A \in \mathcal{A} \quad \text{and} \quad \mu(A) < \delta \implies \int_A f \, \mathrm{d}\mu < \varepsilon.$$

Theorem 8.8 (Finite integral implies uniform integrability). Let $f\colon X \to [0,+\infty]$ be a measurable function. If $\int_X f \,\mathrm{d}\mu < +\infty$, then f is uniformly integrable over measurable sets.

Before proving Theorem 8.8, let us describe the strategy of the proof. Uniform integrability is easier to prove for bounded functions, and in particular for simple functions with finite integral, because their values are bounded by a finite maximum M. For an arbitrary function f with finite integral, we approximate f from below by an increasing sequence of non-negative simple functions $s_n \uparrow f$. By the monotone convergence theorem, the remainder $\int_X (f - s_n) \,\mathrm{d}\mu$ becomes arbitrarily small for large n. Because this approximation error is controlled globally on all of X, the choice of approximating simple function s_n is not dependent on the set A over which we will integrate. Then we use uniform integrability of s_n to choose how small $\mu(A)$ has to be.

Lemma 8.9 (Uniform integrability of a simple function). Every non-negative measurable simple function s with finite integral is uniformly integrable over measurable sets.

Proof of Lemma 8.9. Write $s = \sum_{j=1}^{m} a_j \mathbb{1}_{E_j}$, where $a_1, \dots, a_m \geq 0$ are the distinct values taken by s and $E_j \in \mathcal{A}$ are pairwise disjoint measurable sets. We can ignore the set (if non-empty) on which $s = +\infty$ because this set is μ-null since $\int_X s \,\mathrm{d}\mu < +\infty$.
Set $M \coloneqq \max_{1 \leq j \leq m} a_j$. If $M = 0$, then $s = 0$ identically and $\int_A s \,\mathrm{d}\mu = 0$ for every $A \in \mathcal{A}$. If $M > 0$, then $s(x) \leq M$ for all $x \in X$, and for every $A \in \mathcal{A}$,

$$\int_A s \,\mathrm{d}\mu \leq \int_A M \,\mathrm{d}\mu = M\mu(A).$$

Let $\varepsilon > 0$. Choosing $\delta = \varepsilon / M$, it follows that whenever $\mu(A) < \delta$,

$$\int_A s \,\mathrm{d}\mu \leq M\mu(A) < M\delta = \varepsilon.$$

This proves that s is uniformly integrable over measurable sets. □

We are now ready to prove the full theorem by approximation.

Proof of Theorem 8.8. Step 1: global approximation by simple functions.
Let $\varepsilon > 0$. Choose a sequence of non-negative measurable simple functions $s_n \uparrow f$ pointwise on X. The monotone convergence theorem gives

$$\lim_{n \to \infty} \int_X s_n \,\mathrm{d}\mu = \int_X f \,\mathrm{d}\mu.$$

Since $\int_X f \,\mathrm{d}\mu < +\infty$, all integrals involved are finite. Because $f = s_n + (f - s_n)$ and $s_n \leq f$, positive linearity yields

$$\int_X (f - s_n) \,\mathrm{d}\mu = \int_X f \,\mathrm{d}\mu - \int_X s_n \,\mathrm{d}\mu \longrightarrow 0 \quad \text{as } n \to \infty.$$

Fix an index n large enough that

$$\int_X (f - s_n) \,\mathrm{d}\mu < \varepsilon.$$

Crucially, this choice of n depends on ε and f, but not on any subsequent subset A.

Step 2: uniform control on small sets.
Having fixed the simple function s_n, and since its integral is finite, apply Lemma 8.9 to obtain $\delta =$

$\delta(\varepsilon, s_n) > 0$ such that

$$A \in \mathcal{A} \quad \text{and} \quad \mu(A) < \delta \implies \int_A s_n \, \mathrm{d}\mu < \varepsilon.$$

For any measurable set A satisfying $\mu(A) < \delta$, monotonicity of the integral gives

$$\int_A f \, \mathrm{d}\mu = \int_A f \, \mathrm{d}\mu = \int_A s_n \, \mathrm{d}\mu + \int_A (f - s_n) \, \mathrm{d}\mu \leq \int_A s_n \, \mathrm{d}\mu + \int_X (f - s_n) \, \mathrm{d}\mu < \varepsilon + \varepsilon = 2\varepsilon.$$

Because $\varepsilon > 0$ was arbitrary, the quantity 2ε can be made arbitrarily small. Hence f is uniformly integrable over measurable sets. □

8.4 Integrable signed functions

The Lebesgue integral has thus far been defined only for functions taking values in $[0, +\infty]$. To extend the theory to functions taking both positive and negative values, we decompose a function $f\colon X \to [-\infty, +\infty]$ into a difference of two non-negative functions.

Definition 8.10 (Positive and negative parts). For a measurable function $f\colon X \to [-\infty, +\infty]$, its **positive part** f^+ and **negative part** f^- are defined by

$$f^+(x) \coloneqq \max\{f(x), 0\} = \frac{|f(x)| + f(x)}{2}, \qquad f^-(x) \coloneqq \max\{-f(x), 0\} = \frac{|f(x)| - f(x)}{2}.$$

Then f^+ and f^- are non-negative measurable functions satisfying

$$f = f^+ - f^- \qquad \text{and} \qquad |f| = f^+ + f^-.$$

Equivalently, in terms of indicator functions,

$$f^+ = |f| \mathbb{1}_{\{f \geq 0\}}, \qquad f^- = |f| \mathbb{1}_{\{f < 0\}}.$$

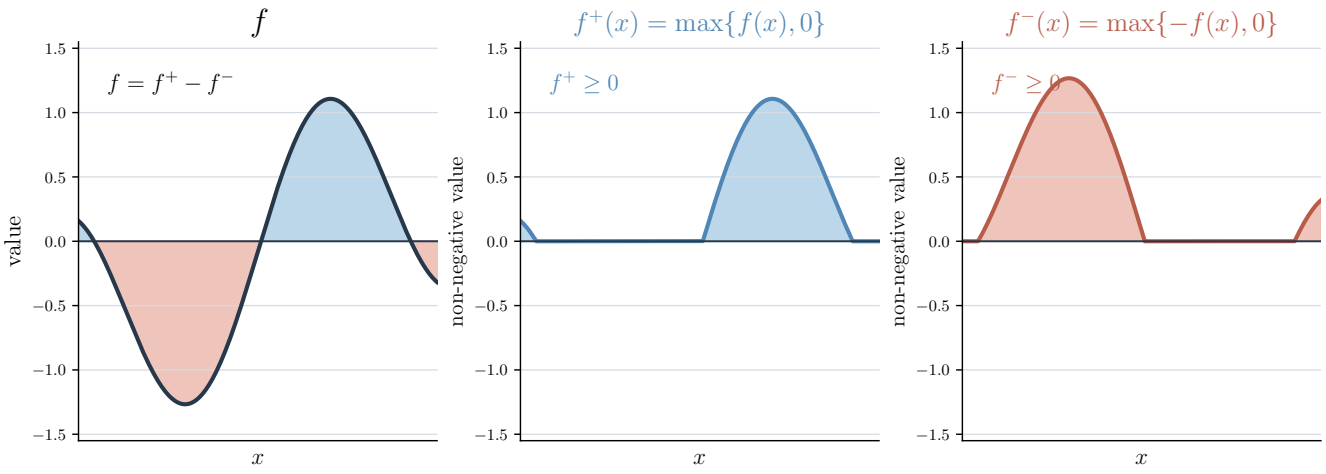


Figure 8.2: The positive and negative parts retain the positive and negative contributions of f as non-negative functions: $f = f^+ - f^-$.

Figure description: Three plots show a signed function f, its non-negative part f plus retaining the portions above the axis, and its non-negative part f minus formed from the magnitudes of the portions below the axis.

Definition 8.11 (Integrable signed function and its integral). A measurable function $f\colon X \to [-\infty, +\infty]$ is **integrable** (or μ-integrable) if

$$\int_X |f| \, \mathrm{d}\mu < +\infty.$$

For an integrable function f, its **Lebesgue integral** is defined by

$$\int_X f \, \mathrm{d}\mu \coloneqq \int_X f^+ \, \mathrm{d}\mu - \int_X f^- \, \mathrm{d}\mu. \tag{8.2}$$

Because $0 \le f^+ \le |f|$ and $0 \le f^- \le |f|$, the integrability condition $\int_X |f| \, \mathrm{d}\mu < +\infty$ guarantees that both non-negative integrals $\int_X f^+ \, \mathrm{d}\mu$ and $\int_X f^- \, \mathrm{d}\mu$ are finite real numbers. Consequently, their difference in (8.2) is a well-defined real number and never produces the indeterminate expression $(+\infty) - (+\infty)$. Moreover, Corollary 8.4 applied to $|f|$ shows that $|f(x)| < +\infty$ for almost every $x \in X$; an integrable signed function is therefore finite-valued almost everywhere.

Proposition 8.12 (Linearity of the signed integral). If f and g are integrable signed functions and $a, b \in \mathbb{R}$, then $af + bg$ is integrable and

$$\int_X (af + bg) \, \mathrm{d}\mu = a \int_X f \, \mathrm{d}\mu + b \int_X g \, \mathrm{d}\mu. \tag{8.3}$$

Proof of Proposition 8.12. The pointwise estimate $|af + bg| \le |a||f| + |b||g|$ and positive linearity show that $\int_X |af + bg| \, \mathrm{d}\mu \le |a| \int_X |f| \, \mathrm{d}\mu + |b| \int_X |g| \, \mathrm{d}\mu < +\infty$, so $af + bg$ is integrable.

Step 1: integration of general differences of non-negative functions.
Suppose $u, v\colon X \to [0, +\infty]$ are non-negative, integrable functions, and set $h \coloneqq u - v$. The pointwise decomposition $h = h^+ - h^-$ gives $h^+ - h^- = u - v$, or equivalently,

$$h^+ + v = h^- + u.$$

Both sides are sums of non-negative measurable functions. Applying positive linearity of the non-negative integral yields

$$\int_X h^+ \, \mathrm{d}\mu + \int_X v \, \mathrm{d}\mu = \int_X h^- \, \mathrm{d}\mu + \int_X u \, \mathrm{d}\mu.$$

Because u and v are integrable, and $0 \le h^+, h^- \le |h| \le u + v$, all four integrals are finite real numbers. Subtracting $\int_X h^- \, \mathrm{d}\mu$ and $\int_X v \, \mathrm{d}\mu$ gives

$$\int_X h \, \mathrm{d}\mu = \int_X h^+ \, \mathrm{d}\mu - \int_X h^- \, \mathrm{d}\mu = \int_X u \, \mathrm{d}\mu - \int_X v \, \mathrm{d}\mu.$$

Step 2: linearity for linear combinations.
For any real numbers $a, b \in \mathbb{R}$, decompose $a = a^+ - a^-$ and $b = b^+ - b^-$. Expanding $af + bg = a(f^+ - f^-) + b(g^+ - g^-)$, we can write

$$af + bg = u - v,$$

where

$$u \coloneqq a^+ f^+ + a^- f^- + b^+ g^+ + b^- g^- \quad \text{and} \quad v \coloneqq a^+ f^- + a^- f^+ + b^+ g^- + b^- g^+.$$

The functions u and v are non-negative and integrable. Applying the result of Step 1 and positive linearity

yields

$$\begin{aligned}\int_X (af+bg)\,\mathrm{d}\mu &= \int_X u\,\mathrm{d}\mu - \int_X v\,\mathrm{d}\mu \\ &= \Big(a^+\int_X f^+\,\mathrm{d}\mu + a^-\int_X f^-\,\mathrm{d}\mu + b^+\int_X g^+\,\mathrm{d}\mu + b^-\int_X g^-\,\mathrm{d}\mu\Big) \\ &\quad - \Big(a^+\int_X f^-\,\mathrm{d}\mu + a^-\int_X f^+\,\mathrm{d}\mu + b^+\int_X g^-\,\mathrm{d}\mu + b^-\int_X g^+\,\mathrm{d}\mu\Big) \\ &= (a^+-a^-)\left(\int_X f^+\,\mathrm{d}\mu - \int_X f^-\,\mathrm{d}\mu\right) + (b^+-b^-)\left(\int_X g^+\,\mathrm{d}\mu - \int_X g^-\,\mathrm{d}\mu\right) \\ &= a\int_X f\,\mathrm{d}\mu + b\int_X g\,\mathrm{d}\mu.\end{aligned}$$

This proves (8.3). □